

Linear First-Order Equations & the Integrating Factor

ANSWER KEY



Recognizing linear form

- 1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.
Already in standard form: $P(x) = 2x$, $Q(x) = x$.
- 2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.
Yes. Standard form: $y' - \frac{1}{x}y = x^2$, so $P(x) = -\frac{1}{x}$, $Q(x) = x^2$.

Finding the integrating factor

- 3 Find the integrating factor for $y' + 3y = e^x$.
 $\mu(x) = e^{\int 3 dx} = e^{3x}$.
- 4 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.
 $\mu(x) = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2}$.

Solving with the integrating factor

- 5 Solve $y' + 3y = e^x$.
With $\mu = e^{3x}$: $(e^{3x}y)' = e^{4x}$, so $e^{3x}y = \frac{1}{4}e^{4x} + C$, giving $y = \frac{1}{4}e^x + Ce^{-3x}$.
- 6 Solve $y' - \frac{2}{x}y = x$, $x > 0$.
With $\mu = x^{-2}$: $(x^{-2}y)' = x^{-1}$, so $x^{-2}y = \ln x + C$, giving $y = x^2 \ln x + Cx^2$.

Initial value problems

- 7 Solve $y' + y = 2$, $y(0) = 0$.
With $\mu = e^x$: $(e^x y)' = 2e^x$, so $e^x y = 2e^x + C$, giving $y = 2 + Ce^{-x}$. Since $y(0) = 0$, $C = -2$:
 $y = 2 - 2e^{-x}$.
- 8 Solve $y' + 2y = 4x$, $y(0) = 1$.
With $\mu = e^{2x}$: $(e^{2x}y)' = 4xe^{2x}$. By parts, $\int 4xe^{2x} dx = 2xe^{2x} - e^{2x} + C$. So $y = 2x - 1 + Ce^{-2x}$. Since $y(0) = 1$, $C = 2$: $y = 2x - 1 + 2e^{-2x}$.

Application: a filling tank

- 9 A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t , the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.

Standard form $Q' + 0.03Q = 6$. With $\mu = e^{0.03t}$: $(e^{0.03t}Q)' = 6e^{0.03t}$, so $e^{0.03t}Q = 200e^{0.03t} + C$, giving $Q = 200 + Ce^{-0.03t}$. Since $Q(0) = 0$, $C = -200$: $Q(t) = 200(1 - e^{-0.03t})$.